Exercise 1

$f_c = 2,5$ GHz

15 dB of insertion loss at 3,75 GHz

$Z_0 = 50\ \Omega$

$L_i = ?$, $C_i = ?$

For a maximally flat filter

$IL = 10 \log_{10}\left(1 + \left(\frac{\omega}{\omega_c}\right)^{2N}\right)$

$\tilde{\omega} = \frac{f}{f_c} = \frac{3,75}{2,5} = 1,5 \xrightarrow{\text{we need}} IL \geq 15\text{ dB at } \tilde{\omega} = 1,5$

$15 \leq 10 \log_{10}\left(1 + 1,5^{2N}\right) \overset{\text{work with real numbers}}{\Rightarrow} 31,62 \leq 1 + 1,5^{2N} \Rightarrow 30,62 \leq 1,5^{2N}$

If we solve for N: $2N \log_{10}(1,5) \geq \log_{10}(30,62)$ we get that:

$2N(0,176) \geq 1,486 \Rightarrow N \geq 4,22$

N has to be integer, so we use $N = 5$

Since this is a Butterworth filter, we compute the prototype table for $N=5$.

$g_1 = 0,6180$

$g_2 = 1,6180$

$g_3 = 2,0$

$g_4 = 1,6180$

$g_5 = 1,0$

Using a T network (a series of inductors each in parallel with a capacitance) we get:

$\omega_c = 2\pi f_c = 15,708 \cdot 10^3$ rad/s

- For inductors, $L_i = \frac{g_k Z_0}{\omega_c}$

$L_1 = \frac{0,6180 \cdot 50}{15,708 \cdot 10^3} = 1,967$ mH

$L_3 = \frac{2,0 \cdot 50}{15,708 \cdot 10^3} = 6,366$ mH

$$L_S = \frac{0,6180 \cdot 50}{15,708 \cdot 10^3} = 1,367 \ \mu H$$

- For Shunt capacitors, $C_i = \frac{8K}{Z_0 \omega_c}$

$$C_2 = \frac{1,6180}{50 \cdot 15,708 \cdot 10^3} = 2,060 \ pF$$

$$C_4 = \frac{1,6180}{50 \cdot 15,708 \cdot 10^3} = 2,060 \ pF$$

Exercise 2

All the data above and $\varepsilon_r = 2,54$, $h = (1/8)''$

Microstrip width between 0,5 and 15

The first step is to calculate the bounding characteristic impedances for Z_h (inductive line) and S_e (capacitive line).

- For an inductive line the width is small, so:

$$\frac{W}{h} = \frac{0,5}{3,175} = 0,157$$

Since the condition $\frac{W}{h} < 1$ is respected we can from graph that:

$\varepsilon_{eff,h} \approx 1,858$

$Z_h \approx 172,9 \ \Omega$

and the guide wave lenght is:

$$\lambda_{gh} = \frac{c}{f_c \sqrt{\varepsilon_{eff,h}}} \approx 88,05 \ mm$$

- For capacitive lines (W at max 15 mm)

$$\frac{W}{h} = \frac{15}{3,175} = 4,724$$

Since for this tipes of lines $\frac{W}{h} > 1$ we can see that

$\varepsilon_{eff,l} \approx 2,179$

$Z_l \approx 34,9 \ \Omega$

$$\lambda_{gl} = \frac{c}{f_c \sqrt{\varepsilon_{eff,l}}} \approx 81,3 \ mm$$

The final step we evaluate the electrical lenght and the physical lenght of each section.

- Inductive sections are 1, 3, 5

$$\theta_i = \frac{g_i Z_0}{Z_h}$$

$$\theta_1 = \theta_5 = \frac{0{,}6180 \cdot 50}{172{,}3} = 0{,}1787 \text{ rad} \Rightarrow l_1 = l_5 = \frac{\theta_1}{2\pi} \lambda_{gh} = 2{,}50 \text{ mm}$$

$$\theta_3 = \frac{2 \cdot 50}{172{,}3} = 0{,}5784 \text{ rad} \Rightarrow l_3 = \frac{\theta_3}{2\pi} \lambda_{gh} = 8{,}10 \text{ mm}$$

- Capacitive sections are 2, 4

$$\theta_i = \frac{g_i Z_e}{Z_0}$$

$$\theta_2 = \theta_4 = \frac{1{,}6180 \cdot 34{,}9}{50} = 1{,}129 \text{ rad} \Rightarrow l_2 = l_4 = \frac{\theta_2}{2\pi} \lambda_{ge} = 14{,}61 \text{ mm}$$

Exercise 3

$f_c = 2{,}5$ GHz

$N = 5$

0,5 dB ripple response

$Z_0 = 50\ \Omega$

$\varepsilon_r = 2{,}54$, $h = \left(\frac{1}{8}\right)''$ with a fixed lenght for each of $\lambda/8$.

From the Chebyshev ripple low-pass prototype table

$g_1 = 1{,}7058$

$g_2 = 1{,}2296$

$g_3 = 2{,}5408$

$g_4 = 1{,}2296$

$g_5 = 1{,}7058$

Now we calculate the characteristic impedance.

Since the physical structure of each step is $\lambda/8$ the electrical lenght is constant at:

$$\theta = \frac{\pi}{4} = 0{,}7854$$

- For inductive steps:

$$Z_{hi} = \frac{g_i Z_0}{\theta}$$

$$Z_{h_1} = Z_{h_5} = \frac{1,7058 \cdot 50}{0,7854} = 108,6\,\Omega$$

$$Z_{h_3} = \frac{2,5408 \cdot 50}{0,7854} = 161,75\,\Omega$$

- For capacitive steps:

$$Z_{l_i} = \frac{Z_0\,\theta}{g_i}$$

$$\lambda_0 = \frac{c}{f_c} = 120\,mm$$

$$Z_{l_2} = Z_{l_4} = \frac{50 \cdot 0,7854}{1,2236} = 31,94\,\Omega$$

By using the standard formula $\frac{W}{h}$ at a given Z_0 we obtain the width

- Inductive sections

$$W/h \approx 0,647$$

$$W_1 = W_5 = 0,647 \cdot 3,175 = 2,05\,mm$$

Lenght

$$\lambda_{g_1} = \frac{\lambda_0}{\sqrt{\varepsilon_{eff,1}}} = \frac{120}{\sqrt{1,944}} \approx 86,0\,mm$$

$$l_1 = l_5 = \frac{\lambda_{g_1}}{8} = 10,75\,mm$$

$$W/h = 0,198$$

$$W_3 = 0,198 \cdot 3,175 = 0,63\,mm$$

$$\lambda_{g_3} = \frac{\lambda_0}{\sqrt{\varepsilon_{eff,3}}} = \frac{120}{\sqrt{1,868}} = 87,8\,mm$$

$$l_3 = \frac{\lambda_{g_3}}{8} = 10,98\,mm$$

- For capacitive lines

$$W/h = 5,27$$

$$W_2 = W_4 = 5,27 \cdot 3,175 = 16,73\,mm$$

The effective permittivity $\varepsilon_{eff,2} \approx 2,19$

$$\lambda_{g_2} = \frac{\lambda_0}{\sqrt{\varepsilon_{eff,2}}} = \frac{120}{\sqrt{2,19}} \approx 81,1\,mm$$

$$l_2 = l_4 = \frac{\lambda_{g_2}}{8} = 10,14\,mm$$